SOFTWARE TESTING AND AUTOMATION

CONSOLIDATED GROUP TEST REPORT

Test Team

FULL STUDENT ID	NAME	RESPONSIBILITY
2022/ICT/62	B.A.N.R. Bulathsinhala	JMeter performance
2022/ICT/63	G.A.M.S.A. Gunarathna	Postman API
2022/ICT/70	W.A.U.D. Wijayakoon	Postman API
2022/ICT/78	U.G.H.S. Ranasinghe	Selenium UI
2022/ICT/90	T.A. Pragas	Selenium UI

Submission: 02 November 2026

01 Application overview

TaskManager is a task-management application developed in a previous semester by T. A. Pragas (2022/ICT/90). The group tests its registration/login, task creation and management, authenticated REST API, and login-endpoint load behavior.

Application: https://github.com/ArunPragas2001/TaskManager
Testing artifacts: https://github.com/nethsara-rahiru/SQA-ICA03
Full tracker: https://docs.google.com/spreadsheets/d/1xntmKoLF67ZUh8P6DqYbO3rtIwb53AHrxQc20ToV6T4/edit

02 Assignment compliance and execution summary

REQUIRED COMPONENT	EVIDENCE IN THIS REPORT
Five-member group / previous-semester app	Group roster and TaskManager overview
Selenium: registration, login, other feature	2022/ICT/78 and 2022/ICT/90 case records and source paths
Postman: POST / GET / PUT / DELETE + assertions	2022/ICT/63 and 2022/ICT/70 cases and collection paths
JMeter: 10 vs 25 users	Configuration and measured metrics from committed HTML reports
Five test cases per member	All 51 individual case rows documented
Bug reports: 2 UI + 1 API, if genuine	Candidate defects and evidence references; requirements verification flagged
Consolidated submission	All five testers included in one document

STUDENT ID	CASES	PASS	FAIL
2022/ICT/62	8	4	4
2022/ICT/63	11	9	2
2022/ICT/70	7	5	2
2022/ICT/78	19	16	3
2022/ICT/90	6	5	1
TOTAL	51	39	12

The tracker dashboard currently shows 55 cases, whereas the detailed tabs contain 51. The individual row count is used here. Pass rate: 76.5%; failure rate: 23.5%. A failed test is not automatically a distinct application defect.

03 Individual test cases — 2022/ICT/62

Tool: Apache JMeter. Target: POST http://localhost:8000/api/auth/login. Each recorded plan uses a single login endpoint with the number of virtual users stated below. Exact ramp-up settings and detailed request configuration are available in the committed .jmx files.

T62 / 001 | Login load test — 10 virtual users | PASS

Precondition	Login endpoint running; test account exists.
Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 10. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.
Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 0.00% errors.
Recorded outcome	Pass
Evidence	2022ICT62/Test 01/Test Plan*.jmx; results.jtl; HTMLReport/.

T62 / 002 | Login load test — 25 virtual users | PASS

Precondition	Login endpoint running; test account exists.
Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 25. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.
Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 0.00% errors.
Recorded outcome	Pass
Evidence	2022ICT62/Test 02/Test Plan*.jmx; results.jtl; HTMLReport/.

T62 / 003 | Login load test — 100 virtual users | PASS

Precondition	Login endpoint running; test account exists.
Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 100. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.

Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 0.00% errors.
Recorded outcome	Pass
Evidence	2022ICT62/Test 03/Test Plan*.jmx; results.jtl; HTMLReport/.

T62 / 004 | Login load test — 500 virtual users | FAIL

Precondition	Login endpoint running; test account exists.
Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 500. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.
Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 7.50% errors.
Recorded outcome	Fail
Evidence	2022ICT62/Test 04/Test Plan*.jmx; results.jtl; HTMLReport/.

T62 / 005 | Login load test — 250 virtual users | FAIL

Precondition	Login endpoint running; test account exists.
Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 250. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.
Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 3.20% errors.
Recorded outcome	Fail
Evidence	2022ICT62/Test 05/Test Plan*.jmx; results.jtl; HTMLReport/.

T62 / 006 | Login load test — 133 virtual users | PASS

Precondition	Login endpoint running; test account exists.

Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 133. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.
Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 0.00% errors.
Recorded outcome	Pass
Evidence	2022ICT62/Test 06/Test Plan*.jmx; results.jtl; HTMLReport/.

T62 / 007 | Login load test — 135 virtual users | FAIL

Precondition	Login endpoint running; test account exists.
Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 135. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.
Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 1.48% errors.
Recorded outcome	Fail
Evidence	2022ICT62/Test 07/Test Plan*.jmx; results.jtl; HTMLReport/.

T62 / 008 | Login load test — 134 virtual users | FAIL

Precondition	Login endpoint running; test account exists.
Test steps	1. Configure JMeter Test Plan. 2. Set thread count to 134. 3. Configure ramp-up and loop count (recorded as one loop). 4. Run the login HTTP request. 5. Review error percentage.
Test data	POST /api/auth/login email: john@example.com password: password123
Expected result	0.00% request errors (tester expectation).
Actual result	Tester tracker reports 0.75% errors.
Recorded outcome	Fail
Evidence	2022ICT62/Test 08/Test Plan*.jmx; results.jtl; HTMLReport/.

04 Individual test cases — 2022/ICT/63

T63 / 001 | Register user | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	POST /api/auth/register with valid name, email and password.
Test data	Samudini Gunarathna; valid email; password SamuAma@1234
Expected result	201 or 200; account created.
Actual result	201; account created.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 002 | Login with unregistered email | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	POST login with email not in database.
Test data	samudiniama@gmail.com; password SamuAma@1234
Expected result	401; invalid credentials.
Actual result	401; invalid email or password.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 003 | Login with invalid email format | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	POST login with invalid email format and JWT header.
Test data	samudiniama&gmail.com; password SamuAma@1234
Expected result	401; invalid credentials.
Actual result	401; invalid email or password.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots.

T63 / 004 | Create task with JWT | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	POST task with valid bearer token.
Test data	Complete Project Report; 2026-11-10; Pending; High
Expected result	201; task created.
Actual result	201; task created.

Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 005 | Create task without title | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	POST task with empty title and valid JWT.
Test data	title empty; 2026-11-10; Pending; High
Expected result	400 validation error.
Actual result	400; title required.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 006 | Create task with past deadline | FAIL

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	POST task with valid JWT and overdue date.
Test data	Review pull requests; 2026-10-05; In Progress; Medium
Expected result	400; reject past date (tester expectation).
Actual result	201; task created.
Recorded outcome	Fail
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 007 | Read protected route without JWT | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	GET /api/tasks without authorization token.
Test data	Authorization header omitted
Expected result	401 unauthorized.
Actual result	401; no token.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 008 | Update task without JWT | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	PUT task deadline without bearer token.
Test data	Review pull requests; deadline 2026-12-05
Expected result	401 unauthorized.

Actual result	401; no token.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 009 | Delete task without JWT | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	DELETE a task without bearer token.
Test data	Existing task; no token
Expected result	401 unauthorized.
Actual result	401; no token.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 010 | Delete task with JWT | PASS

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	DELETE an existing task with valid JWT.
Test data	Existing task; valid JWT
Expected result	200; task removed.
Actual result	200; reported deletion success.
Recorded outcome	Pass
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

T63 / 011 | Excessively long task title | FAIL

Precondition	API server running; required user/task state and token prepared where applicable.
Test steps	POST task with long title and valid JWT.
Test data	Long title; 2026-11-15; Pending; Low
Expected result	400; title length validation.
Actual result	201; task created.
Recorded outcome	Fail
Evidence	2022ICT63/POSTMAN Automation Collection/ and Screenshots/.

05 Individual test cases — 2022/ICT/70

T70 / 001 | Wrong password login | PASS

Precondition	API server running; relevant user and task available.
Test steps	POST login with registered email and incorrect password.
Test data	uthsara@gmail.com; WrongPassword123
Expected result	401; invalid credentials; no token.
Actual result	401; invalid email or password.
Recorded outcome	Pass
Evidence	2022ICT70/Postman_Automation_Collection/ and Testing_automation_screenshots/.

T70 / 002 | Empty login fields | FAIL

Precondition	API server running; relevant user and task available.
Test steps	POST login with email and password empty.
Test data	email empty; password empty
Expected result	400; required fields error.
Actual result	401; invalid email or password.
Recorded outcome	Fail
Evidence	2022ICT70/Postman_Automation_Collection/ and Testing_automation_screenshots/.

T70 / 003 | Valid login and JWT generation | FAIL

Precondition	API server running; relevant user and task available.
Test steps	POST login with registered credentials; verify token and user identity.
Test data	uthsara@gmail.com; Test123456
Expected result	200; JWT and matching user identity.
Actual result	200 and JWT returned; assertion expected a different user's identity.
Recorded outcome	Fail
Evidence	2022ICT70/Postman_Automation_Collection/ and Testing_automation_screenshots/.

T70 / 004 | Read tasks with valid JWT | PASS

Precondition	API server running; relevant user and task available.
Test steps	GET /api/tasks using saved bearer token.
Test data	Authorization: Bearer {{token}}
Expected result	200; array; owned tasks.
Actual result	200; array and ownership checks reported.

Recorded outcome	Pass
Evidence	2022ICT70/Postman_Automation_Collection/ and Testing_automation_screenshots/.

T70 / 005 | Read tasks with invalid JWT | PASS

Precondition	API server running; relevant user and task available.
Test steps	GET /api/tasks with malformed token.
Test data	Bearer abc.def.ghi
Expected result	401; reject invalid token.
Actual result	401; access denied.
Recorded outcome	Pass
Evidence	2022ICT70/Postman_Automation_Collection/ and Testing_automation_screenshots/.

T70 / 006 | Update task with valid JWT | PASS

Precondition	API server running; relevant user and task available.
Test steps	PUT /api/tasks/:id with updated title/status and valid JWT.
Test data	Updated Task - Postman Testing; In Progress; High
Expected result	200; updated fields returned.
Actual result	200; updated fields returned.
Recorded outcome	Pass
Evidence	2022ICT70/Postman_Automation_Collection/ and Testing_automation_screenshots/.

T70 / 007 | Update task with invalid JWT | PASS

Precondition	API server running; relevant user and task available.
Test steps	PUT /api/tasks/:id with malformed JWT.
Test data	Bearer abc.def.ghi; update payload
Expected result	401; update blocked.
Actual result	401; update blocked.
Recorded outcome	Pass
Evidence	2022ICT70/Postman_Automation_Collection/ and Testing_automation_screenshots/.

06 Individual test cases — 2022/ICT/78

T78 / 001 | Registration email without @ | PASS

Precondition	Registration page open.
Test steps	Enter name, malformed email, valid password; submit.
Test data	Harshi; harshigmail.com; Test123
Expected result	Validation error; no account created.
Actual result	Browser displayed missing-@ validation message.
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC001_EmailValidationWithoutAtSign.png

T78 / 002 | Registration email without .com | PASS

Precondition	Registration page open.
Test steps	Enter name, harshi@gmail and password; submit.
Test data	Harshi; harshi@gmail; Test123
Expected result	Application rejects email.
Actual result	Application validation rejected email; remained on registration page.
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC002_EmailValidationTestWithout_com.png

T78 / 003-01 | Password length boundary: 5 | PASS

Precondition	Registration page open; unique email required for each successful registration.
Test steps	1. Open registration form. 2. Enter valid name and email. 3. Enter password of stated length. 4. Submit and observe outcome.
Test data	Password length: 5 characters
Expected result	Reject registration
Actual result	Rejected registration
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC003_PasswordLengthBoundaryTest.png

T78 / 003-02 | Password length boundary: 6 | PASS

Precondition	Registration page open; unique email required for each successful registration.
Test steps	1. Open registration form. 2. Enter valid name and email. 3. Enter password of stated length. 4. Submit and observe outcome.

Test data	Password length: 6 characters
Expected result	Accept registration
Actual result	Accepted registration
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC003_PasswordLengthBoundaryTest.png

T78 / 003-03 | Password length boundary: 7 | PASS

Precondition	Registration page open; unique email required for each successful registration.
Test steps	1. Open registration form. 2. Enter valid name and email. 3. Enter password of stated length. 4. Submit and observe outcome.
Test data	Password length: 7 characters
Expected result	Accept registration
Actual result	Accepted registration
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC003_PasswordLengthBoundaryTest.png

T78 / 003-04 | Password length boundary: 11 | PASS

Precondition	Registration page open; unique email required for each successful registration.
Test steps	1. Open registration form. 2. Enter valid name and email. 3. Enter password of stated length. 4. Submit and observe outcome.
Test data	Password length: 11 characters
Expected result	Accept registration
Actual result	Accepted registration
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC003_PasswordLengthBoundaryTest.png

T78 / 003-05 | Password length boundary: 12 | PASS

Precondition	Registration page open; unique email required for each successful registration.
Test steps	1. Open registration form. 2. Enter valid name and email. 3. Enter password of stated length. 4. Submit and observe outcome.
Test data	Password length: 12 characters
Expected result	Accept registration
Actual result	Accepted registration

Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC003_PasswordLengthBoundaryTest.png

T78 / 003-06 | Password length boundary: 13 | FAIL

Precondition	Registration page open; unique email required for each successful registration.
Test steps	1. Open registration form. 2. Enter valid name and email. 3. Enter password of stated length. 4. Submit and observe outcome.
Test data	Password length: 13 characters
Expected result	Reject registration
Actual result	Accepted registration
Recorded outcome	Fail
Evidence	2022ICT78/Screen Shots/TC003_PasswordLengthBoundaryTest.png

T78 / 004-01 | Task-title length boundary: 2 | FAIL

Precondition	Registered user logged in; dashboard open.
Test steps	1. Open dashboard. 2. Enter task title of stated length. 3. Submit task form. 4. Observe acceptance or rejection.
Test data	Task title length: 2 characters
Expected result	Reject task creation
Actual result	Task creation accepted
Recorded outcome	Fail
Evidence	2022ICT78/Screen Shots/TC004_TaskNameLengthBoundaryTest.png

T78 / 004-02 | Task-title length boundary: 3 | PASS

Precondition	Registered user logged in; dashboard open.
Test steps	1. Open dashboard. 2. Enter task title of stated length. 3. Submit task form. 4. Observe acceptance or rejection.
Test data	Task title length: 3 characters
Expected result	Accept task creation
Actual result	Task creation accepted
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC004_TaskNameLengthBoundaryTest.png

T78 / 004-03 | Task-title length boundary: 4 | PASS

Precondition	Registered user logged in; dashboard open.
Test steps	1. Open dashboard. 2. Enter task title of stated length. 3. Submit task form. 4. Observe acceptance or rejection.
Test data	Task title length: 4 characters
Expected result	Accept task creation
Actual result	Task creation accepted
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC004_TaskNameLengthBoundaryTest.png

T78 / 004-04 | Task-title length boundary: 49 | PASS

Precondition	Registered user logged in; dashboard open.
Test steps	1. Open dashboard. 2. Enter task title of stated length. 3. Submit task form. 4. Observe acceptance or rejection.
Test data	Task title length: 49 characters
Expected result	Accept task creation
Actual result	Task creation accepted
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC004_TaskNameLengthBoundaryTest.png

T78 / 004-05 | Task-title length boundary: 50 | PASS

Precondition	Registered user logged in; dashboard open.
Test steps	1. Open dashboard. 2. Enter task title of stated length. 3. Submit task form. 4. Observe acceptance or rejection.
Test data	Task title length: 50 characters
Expected result	Accept task creation
Actual result	Task creation accepted
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC004_TaskNameLengthBoundaryTest.png

T78 / 004-06 | Task-title length boundary: 51 | FAIL

Precondition	Registered user logged in; dashboard open.

Test steps	1. Open dashboard. 2. Enter task title of stated length. 3. Submit task form. 4. Observe acceptance or rejection.
Test data	Task title length: 51 characters
Expected result	Reject task creation
Actual result	Task creation accepted
Recorded outcome	Fail
Evidence	2022ICT78/Screen Shots/TC004_TaskNameLengthBoundaryTest.png

T78 / 005 | Move task status | PASS

Precondition	Registered user logged in; application running.
Test steps	Create Pending task; click Move to In Progress; check next action.
Test data	Selenium task; Pending; Medium
Expected result	Task moves to In Progress and next action changes.
Actual result	Task moved and next action displayed.
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC005_*.png

T78 / 006 | Edit task | PASS

Precondition	Registered user logged in; application running.
Test steps	Create task; click Edit; change title and priority; save; refresh.
Test data	Edited Selenium task; High
Expected result	Changes persist after refresh.
Actual result	Title and priority persisted.
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC006_*.png

T78 / 007 | Delete task | PASS

Precondition	Registered user logged in; application running.
Test steps	Create task; click Delete; confirm; refresh.
Test data	Existing Selenium task
Expected result	Task removed and remains absent.
Actual result	Task removed; absent after refresh.
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC007_*.png

T78 / 008 | Status column display | PASS

Precondition	Registered user logged in; application running.
Test steps	Create Pending, In Progress and Completed tasks; inspect dashboard.
Test data	Three tasks with distinct statuses
Expected result	Each appears in corresponding column.
Actual result	All three appeared in correct columns.
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC008_*.png

T78 / 009 | Cancel task creation | PASS

Precondition	Registered user logged in; application running.
Test steps	Open Add Task; enter details; cancel; refresh.
Test data	Cancelled Selenium Task; Pending
Expected result	Task not saved or displayed.
Actual result	Task absent after cancellation and refresh.
Recorded outcome	Pass
Evidence	2022ICT78/Screen Shots/TC009_*.png

07 Individual test cases — 2022/ICT/90

T90 / 001 | Login with valid credentials | PASS

Precondition	Taskium running; logged-in user where task operations require it.
Test steps	Open login; enter valid email/password; submit.
Test data	abcd@gmail.com; valid password
Expected result	Dashboard opens.
Actual result	Login succeeded; dashboard opened.
Recorded outcome	Pass
Evidence	2022ICT90/TEST01/; 2022ICT90/TEST01/Login Correct Email, Password.png

T90 / 002 | Login with incorrect password | PASS

Precondition	Taskium running; logged-in user where task operations require it.
Test steps	Open login; enter registered email and wrong password; submit.
Test data	abcd@gmail.com; incorrect password
Expected result	Login rejected with error.
Actual result	Login rejected with error.
Recorded outcome	Pass
Evidence	2022ICT90/TEST02/; 2022ICT90/TEST02/WrongPassword.png

T90 / 003 | Login with unregistered email | PASS

Precondition	Taskium running; logged-in user where task operations require it.
Test steps	Open login; enter unknown email and password; submit.
Test data	unknown_user_987654@example.com; TestPassword123!
Expected result	Login rejected.
Actual result	Login rejected.
Recorded outcome	Pass
Evidence	2022ICT90/TEST03/; 2022ICT90/TEST03/UnRegisteredEmail.png

T90 / 004 | Create task with past deadline | FAIL

Precondition	Taskium running; logged-in user where task operations require it.
Test steps	Open Add Task; enter title, past deadline, priority/status; submit.
Test data	Past Deadline Test Task; 01/01/2020 10:00 AM; High; Pending
Expected result	Task rejected.
Actual result	Task accepted; HTTP 201.

Recorded outcome	Fail
Evidence	2022ICT90/TEST04/; 2022ICT90/TEST04/PastDueDateTest.png

T90 / 005 | Create task with empty title | PASS

Precondition	Taskium running; logged-in user where task operations require it.
Test steps	Open Add Task; leave title empty; enter deadline and options; submit.
Test data	Empty title; 01/01/2030 10:00 AM; High; Pending
Expected result	Submission blocked or rejected.
Actual result	Empty title rejected.
Recorded outcome	Pass
Evidence	2022ICT90/TEST05/; 2022ICT90/TEST05/EmptyTaskTitleTest.png

T90 / 006 | Create multilingual task | PASS

Precondition	Taskium running; logged-in user where task operations require it.
Test steps	Open Add Task; enter multilingual title; save; verify display.
Test data	Team Meeting - Tamil - Sinhala; 01/01/2030; High; Pending
Expected result	Task created; text displays correctly.
Actual result	Task created; multilingual text displayed correctly.
Recorded outcome	Pass
Evidence	2022ICT90/TEST06/; 2022ICT90/TEST06/MultilingualTest.png

08 Bug reports and evidence

The assignment requests attempts to identify at least two UI and one API bug. The following are documented discrepancies, not independently verified defects. Severity and priority are proposed for review.

UI-01 | Past deadline accepted

ATTRIBUTE	DETAIL
Source test	2022/ICT/90 · T90/004
Reproduction steps	Log in; open Add Task; enter a deadline in 2020; submit.
Expected	Past deadline rejected.
Observed	Task created; HTTP 201.
Proposed severity / priority	Medium / Medium
Supporting evidence path	2022ICT90/SCREENSHOTS/AddTaskUsingPastDate.png

Verification note: Confirm the expected behavior against agreed application requirements before treating this candidate as a genuine product defect.

UI-02 | 13-character password accepted

ATTRIBUTE	DETAIL
Source test	2022/ICT/78 · T78/003-06
Reproduction steps	Register with a 13-character password.
Expected	Reject if 12-character maximum is a requirement.
Observed	Registration accepted.
Proposed severity / priority	Medium / Medium
Supporting evidence path	2022ICT78/Screen Shots/TC003_PasswordLengthBoundaryTest.png

Verification note: Confirm the expected behavior against agreed application requirements before treating this candidate as a genuine product defect.

UI-03 | Task title bounds not enforced

ATTRIBUTE	DETAIL
Source test	2022/ICT/78 · T78/004-01, 004-06
Reproduction steps	Create tasks with 2- and 51-character titles.
Expected	Reject if permitted length is 3–50.
Observed	Both accepted.
Proposed severity / priority	Medium / Medium
Supporting evidence path	2022ICT78/Screen Shots/TC004_TaskNameLengthBoundaryTest.png

Verification note: Confirm the expected behavior against agreed application requirements before treating this candidate as a genuine product defect.

API-01 | Empty login input returns 401

ATTRIBUTE	DETAIL
Source test	2022/ICT/70 · T70/002
Reproduction steps	POST /api/auth/login with empty email and password.
Expected	400 Bad Request according to tester's expected contract.
Observed	401 Unauthorized.
Proposed severity / priority	Low / Medium
Supporting evidence path	2022ICT70/Testing_automation_screenshots/TC02_Login_Empty_Fields_1.png

Verification note: Confirm the expected behavior against agreed application requirements before treating this candidate as a genuine product defect.

API-02 | Past deadline accepted via API

ATTRIBUTE	DETAIL
Source test	2022/ICT/63 · T63/006
Reproduction steps	POST a task with deadline 2026-10-05 and valid JWT.
Expected	400 Bad Request if past deadlines are disallowed.
Observed	201 Created.
Proposed severity / priority	Medium / Medium
Supporting evidence path	2022ICT63/Screenshots/With JWT token Create Overdue deadline Task.png

Verification note: Confirm the expected behavior against agreed application requirements before treating this candidate as a genuine product defect.

API-03 | Long task title accepted

ATTRIBUTE	DETAIL
Source test	2022/ICT/63 · T63/011
Reproduction steps	POST a task with a title beyond expected maximum.
Expected	400 if maximum length is specified.
Observed	201 Created.
Proposed severity / priority	Low / Medium
Supporting evidence path	2022ICT63/Screenshots/Check Character Length of the task Name.png

Verification note: Confirm the expected behavior against agreed application requirements before treating this candidate as a genuine product defect.

Not classified as a product defect: 2022/ICT/70 T70/003 — valid login returned 200 and JWT; a Postman identity assertion expected a different test account.

09 Selenium automation evidence

Required scope: registration, login and at least one additional functionality. The combined Selenium coverage includes both registration and login, task creation, task editing, deletion, movement, status rendering and cancellation.

STUDENT ID	AUTOMATION ARTIFACT	EVIDENCE
2022/ICT/78	SeleniumUITestion_TaskManager/pom.xml and src/test/java/tests/*.java	Screen Shots/TC001 through TC009
2022/ICT/90	pom.xml; TEST01 through TEST06/*.java	SCREENSHOTS/ and TEST01–TEST06 PNG files

Execution interpretation: recorded outcomes are taken from the tester sheets. JUnit exit code 0, where reported, indicates testprocesscompletion; assertions and screenshotsshould be reviewed to establish behavior.

10 Postman API automation evidence

CRUD coverage is represented in the 2022/ICT/63 collection: task creation (POST), protected task read (GET), task update (PUT), and task deletion (DELETE). The 2022/ICT/70 collection adds authentication and authorization checks. Response status codes and payloads are documented in the individual test records.

STUDENT ID	COLLECTION LOCATION	RESULT EVIDENCE
2022/ICT/63	2022ICT63/POSTMAN Automation Collection/SQA ICA03 - API Automation.postman_collection.json	POSTMAN Automation Screenshots/ and Screenshots/
2022/ICT/70	2022ICT70/Postman_Automation_Collection/TaskManager-Testing.postman_collection.json	Testing_automation_screenshots/

Automated assertion caveat: 2022/ICT/70 T70/003 contains an inconsistent expected user identity and must be corrected and rerun. The collection files are the authoritative source for the precise assertions.

11 JMeter configuration and measured results

Endpoint: POST /api/auth/login on localhost:8000. Login request uses the recorded test account. JMeter plans, JTL files and HTML dashboards are stored under 2022ICT62/Test 01 through Test 08. The metrics below come from each committed HTMLReport/statistics.json, not the manually entered error percentages.

USERS	SAMPLES	ERROR RATE	MEAN TIME	THROUGHPUT
10	10	0.00%	1,131.5 ms	6.99 req/s
25	25	0.00%	2,826.5 ms	6.16 req/s
100	100	0.00%	6,630.8 ms	7.63 req/s
500	500	7.80%	28,355.5 ms	8.74 req/s
250	250	4.00%	14,897.7 ms	8.42 req/s
133	133	0.00%	9,334.3 ms	7.75 req/s
135	135	0.00%	8,963.1 ms	7.64 req/s
134	134	0.00%	10,425.9 ms	7.00 req/s

Required comparison — 10 versus 25 users

At 10 users: 0.0% errors, 1,131.5 ms mean response, 6.99 requests/s. At 25 users: 0.0% errors, 2,826.5 ms mean response, 6.16 requests/s. Mean response time increased approximately 149.8%; throughput decreased approximately 11.9%. These measurements are specific to the recorded environment.

Load test interpretation

The committed summaries report zero errors at 100, 133, 134 and 135 users; 4.0% at 250 users; and 7.8% at 500 users. The spreadsheet differs from the committed summary for some higher-load tests. This is a documentation

discrepancyrequiringreconciliation, not evidence of anexact concurrencythreshold.

12 Conclusion and pre-submission verification

The recorded team work includes 51 cases across all three mandated testing tools, with 39 passes and 12 failures. All five members have at least five test cases. The repository includes executable test assets and evidence folders. Before submitting: (1) verify the intended password and task-title validation rules; (2) correct and rerun the inconsistent Postman identity assertion; (3) reconcile JMeter spreadsheet error rates with committed JTL/HTML statistics; (4) correct the dashboard totals; (5) ensure the supporting screenshots and test execution reports are

accessible to thelecturer.

Appendix — Evidence access

GitHub testing repository: https://github.com/nethsara-rahiru/SQA-ICA03

Google Sheets tracker:
https://docs.google.com/spreadsheets/d/1xntmKoLF67ZUh8P6DqYbO3rtIwb53AHrxQc20ToV6T4/edit

Original assignment: Software Testing and Automation Assignment (three-page PDF). The lecturer requires one consolidated document including all test cases, bug reports, Selenium results, Postman collection and results, and JMeter configurations and comparisons.